

Module end assignment :

## 1.Data entry, import, and basic organization using tables

Converting all the sheets into tables and giving names

### Customer\_table

|                |                           |                  |                     |                     |                 |                                                 |                                                |                                       |
|----------------|---------------------------|------------------|---------------------|---------------------|-----------------|-------------------------------------------------|------------------------------------------------|---------------------------------------|
| Table Name:    | Summarize with PivotTable | Insert Slicer    | Export              | Refresh             | Open in Browser | Properties                                      | <input checked="" type="checkbox"/> Header Row | <input type="checkbox"/> First Column |
| Customer_table | Remove Duplicates         | Convert to Range | Unlink              | Unlink              | Unlink          | <input type="checkbox"/> Total Row              | <input type="checkbox"/> Last Column           |                                       |
| Resize Table   | Convert to Range          | Tools            | External Table Data | Table Style Options |                 | <input checked="" type="checkbox"/> Banded Rows | <input type="checkbox"/> Banded Columns        |                                       |

| Age | Gender | City              | State          | Country | Loyalty_Level |
|-----|--------|-------------------|----------------|---------|---------------|
| 55  | Male   | Donaldton         | North Carolina | USA     |               |
| 26  | Female | Phillipsmouth     | North Dakota   | USA     | Platinum      |
| 29  | Female | Kylebury          | Alabama        | USA     | Platinum      |
| 59  | Female | Thomastown        | Ohio           | USA     | Bronze        |
| 64  | Female | Nicholasshire     | California     | USA     | Silver        |
| 24  | Male   | East Amandamouth  | Oregon         | USA     | Bronze        |
| 29  | Female | Deborahborough    | South Dakota   | USA     | Platinum      |
| 29  | Female | West Pamelafurt   | West Virginia  | USA     | Bronze        |
| 53  | Female | Maynardton        | New Hampshire  | USA     | Gold          |
| 64  | Male   | Melissashire      | Pennsylvania   | USA     | Platinum      |
| 53  | Male   | Port Michaelburgh | Pennsylvania   | USA     | Bronze        |
| 28  | Female | East Krista       | New York       | USA     | Gold          |
| 55  | Female | North Kathryn     | Kansas         | USA     | Platinum      |
| 32  | Female | New Becky         | Montana        | USA     | Silver        |
| 53  | Female | Lake Kristamouth  | Nebraska       | USA     | Silver        |
| 49  | Female | Lawsonhaven       | Washington     | USA     | Platinum      |
| 24  | Female | South Christopher | Pennsylvania   | USA     | Bronze        |
| 60  | Female | Brandonville      | Georgia        | USA     | Platinum      |
| 43  | Male   | South William     | Kansas         | USA     |               |

### Product\_table

|               |                           |                  |                     |                     |                 |                                                 |                                                 |                                    |
|---------------|---------------------------|------------------|---------------------|---------------------|-----------------|-------------------------------------------------|-------------------------------------------------|------------------------------------|
| Table Name:   | Summarize with PivotTable | Insert Slicer    | Export              | Refresh             | Open in Browser | Properties                                      | <input checked="" type="checkbox"/> Header Row  | <input type="checkbox"/> Total Row |
| Product_table | Remove Duplicates         | Convert to Range | Unlink              | Unlink              | Unlink          | <input type="checkbox"/> Total Row              | <input checked="" type="checkbox"/> Banded Rows |                                    |
| Resize Table  | Convert to Range          | Tools            | External Table Data | Table Style Options |                 | <input checked="" type="checkbox"/> Banded Rows | <input type="checkbox"/> Banded Columns         |                                    |

| Product_ID | Product_Name                | Category    | Sub_Category | Brand                        | Cost   | Stock |
|------------|-----------------------------|-------------|--------------|------------------------------|--------|-------|
| PROD00     | Brother Laser Printer       | Electronics | Computer     | Hunt Inc.                    | 112.42 | 418   |
| PROD01     | Men's Graphic T-Shirt       | Clothing    | Top          | Terry-Moore                  | 342.56 | 181   |
| PROD02     | Women's Casual Hoodie       | Clothing    | Outerwear    | Sims Inc.                    |        | 352   |
| P-3        | Men's Running Jacket        | Clothing    | Outerwear    | Pineda Ltd                   | 264.67 | 452   |
| PROD04     | Women's Yoga Pants          | Clothing    | Bottoms      | Gonzales-Davis               | 306.46 | 487   |
| PROD05     | Men's Leather Belt          | Clothing    | Accessories  | Baldwin, Garza and Gonzalez  | 261    | 59    |
| PROD06     | Herbal Face Moisturizer     | Beauty      | Skinware     | Davidson PLC                 | 312.8  | 356   |
| PROD07     | LED Table Lamp              | Home        | Lighting     | Harrell LLC                  | 218.85 | 170   |
| PROD08     | Wireless Bluetooth Earbuds  | Electronics | Audio        | Perez-Johnston               | 439.07 | 142   |
| PROD09     | Adjustable Dumbbell Set     | Sports      | Fitness      | Will P.C.                    | 499.72 | 323   |
| PROD10     | Ceramic Kitchen Bowl Set    | Home        | Kitchen      | Hamilton Group               | 298.85 | 308   |
| PROD11     | HD Smart TV Stick           | Electronics | Accessories  | Gibson, Hobbs and Weaver     | 449.4  |       |
| PROD12     | Hair Strengthening Shampoo  | Beauty      | Haircare     | Rice LLC                     | 115.62 | 460   |
| PROD13     | USB-C Fast Charger          | Electronics | Accessories  | Dawson-Perkins               | 242.23 | 453   |
| PROD14     | Decorative Wall Mirror      | Home        | Decor        | Luna, Anderson and Bowman    | 48.01  | 439   |
| PROD15     | 4x Home Projector           | Electronics | Appliances   | Thomas LLC                   | 465.34 | 303   |
| P-18       | Beard Grooming Kit          | Beauty      | Grooming     | Erickson-Green               | 337.42 | 256   |
| PROD17     | Soccer Training Cone Set    | Sports      | Bottoms      | Ford, Murphy and Wilson      | 144.53 | 367   |
| PROD18     | Women's Denim Jeans         | Clothing    | Bottoms      | Guerra-Hendrix               |        | 343   |
| PROD19     | Knee Support Brace          | Sports      | Equipment    | May-Reese                    | 21.34  | 65    |
| PROD20     | Outdoor Sports Water Bottle | Sports      | Accessories  | Soto-Ellis                   | 131.51 | 340   |
| PROD21     | Multi-Purpose Storage Box   | Home        | Storage      | Gardner-Montes               | 323.46 | 76    |
| PROD22     | Carbon Fiber Tennis Racket  | Sports      | Equipment    | Flores, Pineda and Rodriguez | 484.62 | 432   |
| PROD23     | Men's Fermal Shirt          | Clothing    | Top          | Moon, Mora and Santos        |        | 360   |
| PROD24     | Foldable Camping Chair      | Sports      | Outdoor      | Cook Ltd                     | 166.06 | 340   |
| PROD25     | Women's Scarf Set           | Clothing    | Accessories  | Brown and Sons               | 174.02 | 268   |
| PROD26     | Wireless Bluetooth Speaker  | Electronics | Audio        | Reed-Schmitt                 | 323.41 | 213   |
| PROD27     | Decorative Floor Vase       | Home        | Decor        | Khan, Jones and Paul         |        | 378   |
| PROD28     | Sports Stopwatch Timer      | Sports      | Accessories  | Burns-Hansen                 | 38.76  | 137   |
| PROD29     | Fabric Sofa Throw Blanket   | Home        | Decor        | Hoffman Group                | 219.48 | 276   |

### Store\_table

|              |                           |                  |                     |                     |                 |                                                 |                                                |                                       |
|--------------|---------------------------|------------------|---------------------|---------------------|-----------------|-------------------------------------------------|------------------------------------------------|---------------------------------------|
| Table Name:  | Summarize with PivotTable | Insert Slicer    | Export              | Refresh             | Open in Browser | Properties                                      | <input checked="" type="checkbox"/> Header Row | <input type="checkbox"/> First Column |
| Store_table  | Remove Duplicates         | Convert to Range | Unlink              | Unlink              | Unlink          | <input type="checkbox"/> Total Row              | <input type="checkbox"/> Last Column           |                                       |
| Resize Table | Convert to Range          | Tools            | External Table Data | Table Style Options |                 | <input checked="" type="checkbox"/> Banded Rows | <input type="checkbox"/> Banded Columns        |                                       |

| Store_ID | Store_Name                      | Region | City               | Store_Type |
|----------|---------------------------------|--------|--------------------|------------|
| STORE0   | Rivera-Warner                   | East   | Washingtonfurt     | Flagship   |
| STORE1   | Martinez, Adkins and Johnson    | North  | Danielmouth        | Flagship   |
| STORE2   | Black-Horn                      | South  | North Nancyview    | Outlet     |
| STORE3   | Johnson-Wright                  | North  | Hoodberg           | Outlet     |
| STORE4   | Alexander Group                 | South  | Port Sharon        | Online     |
| STORE5   | Gray Ltd                        | North  | New Michaelchester | Outlet     |
| STORE6   | Benson-Elliott                  | East   | Port Josephberg    | Online     |
| STORE7   | Martinez-Mullins                | South  | Williamsfurt       | Online     |
| STORE8   | Bean, Jones and Park            | West   | Thompsonborough    | Online     |
| STORE9   | Moore, Padilla and Butler       | South  | East Johnstad      | Flagship   |
| STORE10  | Hill, Wagner and Ballard        | East   | East Davidville    | Online     |
| STORE11  | Aguirre-Keller                  | South  | Lake Michael       | Online     |
| STORE12  | Thomas, Bridges and Harris      | North  | Sabrinatown        | Online     |
| STORE13  | Gardner, Humphrey and Contreras | East   | Johnsonville       | Flagship   |
| STORE14  | Mcdonald Inc                    | South  | Christopherland    | Outlet     |
| STORE15  | Shelton PLC                     | North  | East Jenniferstad  | Flagship   |
| STORE16  | Frederick-Wright                | West   | New Jimmymouth     | Flagship   |
| STORE17  | Holloway Ltd                    | North  | Bensonhaven        | Flagship   |
| STORE18  | Guerrero-Schmitt                | East   | East Stevechester  | Flagship   |

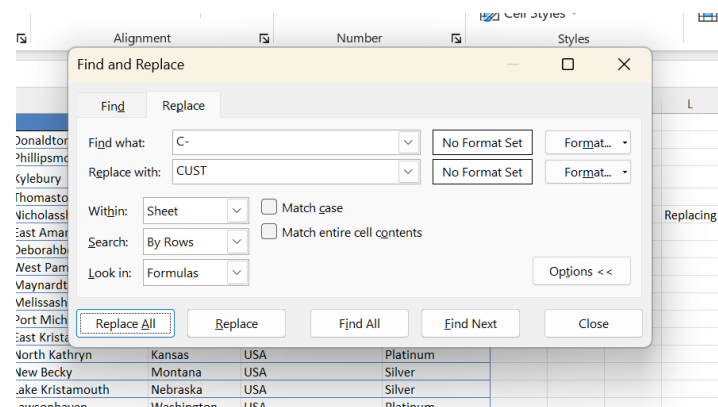
## sales\_table

| Sales_ID | Order_Date | Customer_ID | Product_ID | Store_ID | Quantity | Unit_Price | Discount | Payment_Type | Total_Amount |
|----------|------------|-------------|------------|----------|----------|------------|----------|--------------|--------------|
| SALE0    | 2024-07-07 | CUST1409    | PROD97     | STORE3   | 1        | 218.04     | 0.09     | Credit Card  | 198.4164     |
| SALE1    | 2023-12-25 | CUST1467    | PROD74     | STORE17  | 3        | 104.47     | 0.27     | PayPal       | 228.7893     |
| SALE2    | 2024-06-03 | CUST1056    | PROD49     | STORE8   | 4        | 914.05     | 0.25     | Credit Card  | 2742.15      |
| SALE3    | 2023-11-29 | CUST1346    | PROD76     | STORE8   | 1        | 921.51     | 0.16     | COO          | 774.0684     |
| SALE4    | 2024-01-06 | CUST1493    | PROD59     | STORE5   | 3        | 745.3      | 0.18     | Credit Card  | 1833.438     |
| SALE5    | 2024-07-26 | CUST1116    | PROD15     | STORE7   | 3        | 147.01     | 0.19     | COO          | 357.2343     |
| SALE6    | 2025-07-12 | CUST1016    | PROD56     | STORE4   | 2        | 785.31     | 0.15     | PayPal       | 1335.027     |
| SALE7    | 2023-11-18 | CUST1152    | PROD70     | STORE8   | 4        | 342.39     | 0.23     | PayPal       | 1054.5612    |
| SALE8    | 2025-05-04 | CUST1187    | PROD99     | STORE5   | 2        | 448.25     | 0.07     | PayPal       | 1250.6175    |
| SALE9    | 2025-07-31 | CUST1464    | PROD33     | STORE0   | 2        | 873.17     | 0.23     | PayPal       | 1344.6818    |
| SALE10   | 2024-02-13 | CUST1065    | PROD24     | STORE14  | 1        | 278.63     | 0.07     | Credit Card  | 259.1259     |
| SALE11   | 2025-05-24 | CUST1320    | PROD39     | STORE1   | 4        | 286.92     | 0.17     | Credit Card  | 952.5744     |
| SALE12   | 2024-08-24 | CUST1237    | PROD88     | STORE4   | 4        | 644.58     | 0.01     | Credit Card  | 2552.5368    |
| SALE13   | 2025-08-06 | CUST1443    | PROD34     | STORE16  | 2        | 316.9      | 0.3      | PayPal       | 443.66       |
| SALE14   | 2025-06-21 | CUST1348    | PROD24     | STORE15  | 4        | 403.47     | 0.29     | PayPal       | 1145.8548    |
| SALE15   | 2024-12-10 | CUST1435    | PROD36     | STORE9   | 2        | 871.34     | 0.04     | COO          | 2509.4592    |
| SALE16   | 2025-09-13 | CUST1276    | PROD65     | STORE19  | 2        | 451.98     | 0.25     | Credit Card  | 677.97       |
| SALE17   | 2024-12-11 | CUST1079    | PROD27     | STORE4   | 2        | 502.69     | 0.23     | Credit Card  | 774.1426     |
| SALE18   | 2024-08-13 | CUST1094    | PROD39     | STORE4   | 2        | 559.23     | 0.25     | COO          | 838.845      |

**2.Data cleaning: find/remove duplicates, handle missing/inconsistent data, correct formatting errors, and validate data**

## Customer\_table

For customer\_ID - Replacing the C- with CUST to make it consistent

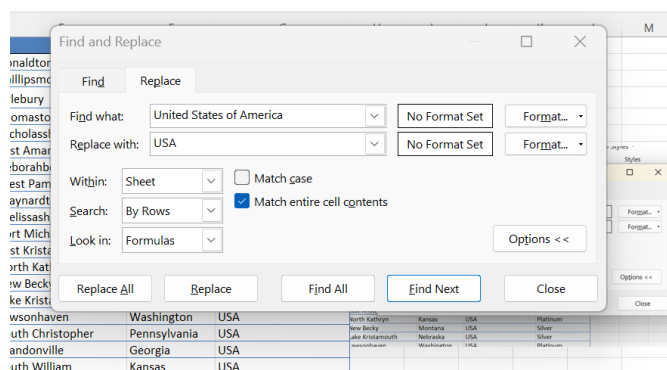


Name - PROPER(TRIM(CLEAN(B2)))

City – PROPER(TRIM(CLEAN(E2)))

State – PROPER(TRIM(CLEAN(F2)))

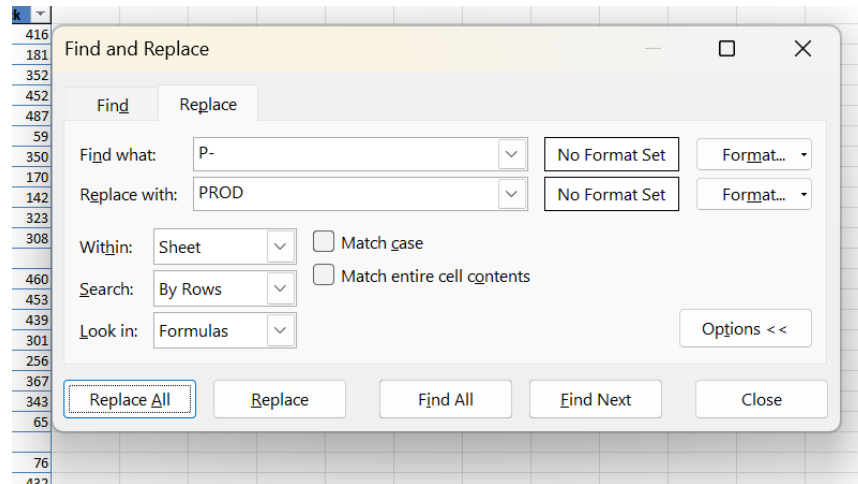
Country – Replacing United States of America to USA



## Product\_table

For Product Id

Replacing P- with PROD



Cost – IF(ISBLANK([@Cost]),AVERAGE(\$F\$2:\$F\$101),[@Cost])

Stock – IF(ISBLANK(G2),AVERAGE(\$G\$2:\$G\$101),G2)

## Store\_table

There is no inconsistency in text , changed the data type of the columns

## Sales\_table

Quantity – IF(ISBLANK([@Quantity]),AVERAGE(\$F\$2:\$F\$2001),[@Quantity])

Unit\_Price – IF(ISBLANK([@Unit\_Price]),AVERAGE(\$G\$2:\$G\$2001),[@Unit\_Price])

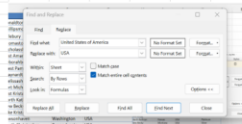
Discount – IF(ISBLANK([@Discount]),AVERAGE(\$H\$2:\$H\$2001),[@Discount])

Total\_amount – [@[Unit\_Price]]\*[@Quantity]-[@Discount]

|                                    |             |            |          |          |            |          |              |              |   |   |   |   |   |   |   |   |   |  |  |  |
|------------------------------------|-------------|------------|----------|----------|------------|----------|--------------|--------------|---|---|---|---|---|---|---|---|---|--|--|--|
| 07-07-2024                         |             |            |          |          |            |          |              |              |   |   |   |   |   |   |   |   |   |  |  |  |
|                                    | C           | D          | E        | F        | G          | H        | I            | J            | K | L | M | N | O | P | Q | R | S |  |  |  |
|                                    | Customer_ID | Product_ID | Store_ID | Quantity | Unit_Price | Discount | Payment_Type | Total_Amount |   |   |   |   |   |   |   |   |   |  |  |  |
| 1                                  | CUST1409    | PROD97     | STORE3   | 1        | 218.04     | 0.09     | Credit Card  | 217.95       |   |   |   |   |   |   |   |   |   |  |  |  |
| 2                                  | CUST1467    | PROD74     | STORE17  | 3        | 104.47     | 0.27     | PayPal       | 313.14       |   |   |   |   |   |   |   |   |   |  |  |  |
| 3                                  | CUST1056    | PROD49     | STORE8   | 4        | 914.05     | 0.25     | Credit Card  | 3655.95      |   |   |   |   |   |   |   |   |   |  |  |  |
| 4                                  | CUST1346    | PROD76     | STORE8   | 1        | 921.51     | 0.16     | COD          | 921.35       |   |   |   |   |   |   |   |   |   |  |  |  |
| 5                                  | CUST1493    | PROD59     | STORE5   | 3        | 745.30     | 0.18     | Credit Card  | 2235.72      |   |   |   |   |   |   |   |   |   |  |  |  |
| 6                                  | CUST1116    | PROD15     | STORE7   | 3        | 147.01     | 0.19     | COD          | 440.84       |   |   |   |   |   |   |   |   |   |  |  |  |
| 7                                  | CUST1016    | PROD56     | STORE4   | 2        | 785.31     | 0.15     | PayPal       | 1570.47      |   |   |   |   |   |   |   |   |   |  |  |  |
| 8                                  | CUST1152    | PROD70     | STORE8   | 4        | 342.39     | 0.23     | PayPal       | 1369.33      |   |   |   |   |   |   |   |   |   |  |  |  |
| 9                                  | CUST1187    | PROD99     | STORE5   | 2        | 448.25     | 0.07     | PayPal       | 1111.68      |   |   |   |   |   |   |   |   |   |  |  |  |
| 10                                 | CUST1464    | PROD33     | STORE0   | 2        | 873.17     | 0.23     | PayPal       | 1746.11      |   |   |   |   |   |   |   |   |   |  |  |  |
| 11                                 | CUST1065    | PROD24     | STORE14  | 1        | 278.63     | 0.07     | Credit Card  | 278.56       |   |   |   |   |   |   |   |   |   |  |  |  |
| 12                                 | CUST1320    | PROD39     | STORE1   | 4        | 286.92     | 0.17     | Credit Card  | 1147.51      |   |   |   |   |   |   |   |   |   |  |  |  |
| 13                                 | CUST1237    | PROD88     | STORE4   | 4        | 644.58     | 0.01     | Credit Card  | 2578.31      |   |   |   |   |   |   |   |   |   |  |  |  |
| 14                                 | CUST1443    | PROD34     | STORE16  | 2        | 316.90     | 0.30     | PayPal       | 633.50       |   |   |   |   |   |   |   |   |   |  |  |  |
| 15                                 | CUST1348    | PROD24     | STORE15  | 4        | 403.47     | 0.29     | PayPal       | 1613.59      |   |   |   |   |   |   |   |   |   |  |  |  |
| 16                                 | CUST1435    | PROD36     | STORE9   | 2        | 871.34     | 0.04     | COD          | 2161.06      |   |   |   |   |   |   |   |   |   |  |  |  |
| 17                                 | CUST1276    | PROD65     | STORE19  | 2        | 451.98     | 0.25     | Credit Card  | 903.71       |   |   |   |   |   |   |   |   |   |  |  |  |
| 18                                 | CUST1079    | PROD27     | STORE4   | 2        | 502.69     | 0.23     | Credit Card  | 1005.15      |   |   |   |   |   |   |   |   |   |  |  |  |
| 19                                 | CUST1094    | PROD39     | STORE4   | 2        | 559.23     | 0.25     | COD          | 1118.21      |   |   |   |   |   |   |   |   |   |  |  |  |
| 20                                 | CUST1239    | PROD60     | STORE13  | 4        | 513.98     | 0.04     | COD          | 2055.98      |   |   |   |   |   |   |   |   |   |  |  |  |
| 21                                 | CUST1144    | PROD72     | STORE17  | 3        | 986.04     | 0.27     | COD          | 2957.85      |   |   |   |   |   |   |   |   |   |  |  |  |
| 22                                 | CUST1396    | PROD43     | STORE18  | 1        | 589.56     | 0.22     | Credit Card  | 589.34       |   |   |   |   |   |   |   |   |   |  |  |  |
| merged table                       |             |            |          |          |            |          |              |              |   |   |   |   |   |   |   |   |   |  |  |  |
| Customer_Dim                       |             |            |          |          |            |          |              |              |   |   |   |   |   |   |   |   |   |  |  |  |
| Product_Dim                        |             |            |          |          |            |          |              |              |   |   |   |   |   |   |   |   |   |  |  |  |
| Store_Dim                          |             |            |          |          |            |          |              |              |   |   |   |   |   |   |   |   |   |  |  |  |
| Sales_Fact                         |             |            |          |          |            |          |              |              |   |   |   |   |   |   |   |   |   |  |  |  |
| Calculation Descriptive Statistics |             |            |          |          |            |          |              |              |   |   |   |   |   |   |   |   |   |  |  |  |

Quantity IF(ISBLANK([@Quantity]),AVERAGE(\$F\$2:\$F\$2001),[@Quantity])  
Unit\_Price IF(ISBLANK([@Unit\_Price]),AVERAGE(\$G\$2:\$G\$2001),[@Unit\_Price])  
Discount IF(ISBLANK([@Discount]),AVERAGE(\$H\$2:\$H\$2001),[@Discount])  
Total\_amount [[@Unit\_Price]]\*[@Quantity]-[@Discount]

| G18 |         |                                 |       |                    |          |   |  |  |  |  |  |  |  |  |  |
|-----|---------|---------------------------------|-------|--------------------|----------|---|--|--|--|--|--|--|--|--|--|
|     | A       | B                               | C     | D                  | E        | F |  |  |  |  |  |  |  |  |  |
| 2   | STORE0  | Rivera-Warner                   | East  | Washingtonfurt     | Flagship |   |  |  |  |  |  |  |  |  |  |
| 3   | STORE1  | Martinez, Adkins and Johnson    | North | Danielmouth        | Flagship |   |  |  |  |  |  |  |  |  |  |
| 4   | STORE2  | Black-Horn                      | South | North Nancyview    | Outlet   |   |  |  |  |  |  |  |  |  |  |
| 5   | STORE3  | Johnson-Wright                  | North | Hoodberg           | Outlet   |   |  |  |  |  |  |  |  |  |  |
| 6   | STORE4  | Alexander Group                 | South | Port Sharon        | Online   |   |  |  |  |  |  |  |  |  |  |
| 7   | STORE5  | Gray Ltd                        | North | New Michaelchester | Outlet   |   |  |  |  |  |  |  |  |  |  |
| 8   | STORE6  | Benson-Elliott                  | East  | Port Josephberg    | Online   |   |  |  |  |  |  |  |  |  |  |
| 9   | STORE7  | Martinez-Mullins                | South | Williamsfurt       | Online   |   |  |  |  |  |  |  |  |  |  |
| 10  | STORE8  | Bean, Jones and Park            | West  | Thompsonborough    | Online   |   |  |  |  |  |  |  |  |  |  |
| 11  | STORE9  | Moore, Padilla and Butler       | South | East Johnstad      | Flagship |   |  |  |  |  |  |  |  |  |  |
| 12  | STORE10 | Hill, Wagner and Ballard        | East  | East Davidville    | Online   |   |  |  |  |  |  |  |  |  |  |
| 13  | STORE11 | Aguirre-Keller                  | South | Lake Michael       | Online   |   |  |  |  |  |  |  |  |  |  |
| 14  | STORE12 | Thomas, Bridges and Harris      | North | Sabrinatown        | Online   |   |  |  |  |  |  |  |  |  |  |
| 15  | STORE13 | Gardner, Humphrey and Contreras | East  | Johnsonville       | Flagship |   |  |  |  |  |  |  |  |  |  |
| 16  | STORE14 | Mcdonald Inc                    | South | Christopherland    | Outlet   |   |  |  |  |  |  |  |  |  |  |
| 17  | STORE15 | Shelton PLC                     | North | East Jenniferstad  | Flagship |   |  |  |  |  |  |  |  |  |  |
| 18  | STORE16 | Frederick-Wright                | West  | New Jimmymouth     | Flagship |   |  |  |  |  |  |  |  |  |  |
| 19  | STORE17 | Holloway Ltd                    | North | Bensonhaven        | Flagship |   |  |  |  |  |  |  |  |  |  |
| 20  | STORE18 | Guerrero-Schmitt                | East  | East Stevechester  | Flagship |   |  |  |  |  |  |  |  |  |  |
| 21  | STORE19 | Mann-Austin                     | East  | Vasquezborough     | Online   |   |  |  |  |  |  |  |  |  |  |



|    | Product_ID | Product_Name                | Category    | Sub_Category | Brand                        | Cost   | Stock |  |  |  |  |  |  |  |  |
|----|------------|-----------------------------|-------------|--------------|------------------------------|--------|-------|--|--|--|--|--|--|--|--|
| 1  | PROD00     | Brother Laser Printer       | Electronics | Computer     | Hunt Inc                     | 112.42 | 416   |  |  |  |  |  |  |  |  |
| 2  | PROD01     | Men's Graphic T-shirt       | Clothing    | Tops         | Terry-Moore                  | 342.56 | 181   |  |  |  |  |  |  |  |  |
| 3  | PROD02     | Women's Casual Hoodie       | Clothing    | Outerwear    | Sims Inc                     | 254.04 | 352   |  |  |  |  |  |  |  |  |
| 4  | PROD03     | Men's Running Jacket        | Clothing    | Outerwear    | Pineda Ltd                   | 364.67 | 452   |  |  |  |  |  |  |  |  |
| 5  | PROD04     | Women's Yoga Pants          | Clothing    | Bottoms      | Gonzales-Davis               | 306.46 | 487   |  |  |  |  |  |  |  |  |
| 6  | PROD05     | Men's Leather Belt          | Clothing    | Accessories  | Baldwin, Garza and Gonzalez  | 261.00 | 59    |  |  |  |  |  |  |  |  |
| 7  | PROD06     | Herbal Face Moisturizer     | Beauty      | Skincare     | Davidson LLC                 | 312.80 | 350   |  |  |  |  |  |  |  |  |
| 8  | PROD07     | LED Table Lamp              | Home        | Lighting     | Harrell LLC                  | 218.86 | 170   |  |  |  |  |  |  |  |  |
| 9  | PROD08     | Wireless Bluetooth Earbuds  | Electronics | Audio        | Perez-Johnston               | 439.07 | 142   |  |  |  |  |  |  |  |  |
| 10 | PROD09     | Adjustable Dumbbell Set     | Sports      | Fitness      | Hill PLC                     | 495.72 | 323   |  |  |  |  |  |  |  |  |
| 11 | PROD10     | Ceramic Kitchen Bowl Set    | Home        | Kitchen      | Hamilton Group               | 296.86 | 308   |  |  |  |  |  |  |  |  |
| 12 | PROD11     | HD Smart TV 55in            | Electronics | Accessories  | Gibson, Hobbs and Weaver     | 449.40 | 273   |  |  |  |  |  |  |  |  |
| 13 | PROD12     | Hair Strengthening Shampoo  | Beauty      | Haircare     | Rice LLC                     | 115.82 | 460   |  |  |  |  |  |  |  |  |
| 14 | PROD13     | USB-C Fast Charger          | Electronics | Accessories  | Dawson-Perkins               | 242.23 | 453   |  |  |  |  |  |  |  |  |
| 15 | PROD14     | Decorative Wall Mirror      | Home        | Decor        | Luna, Anderson and Bowman    | 48.01  | 439   |  |  |  |  |  |  |  |  |
| 16 | PROD15     | 4K Home Projector           | Electronics | Appliances   | Thomas LLC                   | 465.34 | 301   |  |  |  |  |  |  |  |  |
| 17 | PROD16     | Beard Grooming Kit          | Beauty      | Grooming     | Erickson-Green               | 337.42 | 256   |  |  |  |  |  |  |  |  |
| 18 | PROD17     | Soccer Training Cone Set    | Sports      | Accessories  | Ford, Murphy and Wilson      | 144.53 | 367   |  |  |  |  |  |  |  |  |
| 19 | PROD18     | Women's Denim Jeans         | Clothing    | Bottoms      | Guerra-Hendrix               | 254.04 | 343   |  |  |  |  |  |  |  |  |
| 20 | PROD19     | Knee Support Brace          | Sports      | Equipment    | May-Resse                    | 21.34  | 65    |  |  |  |  |  |  |  |  |
| 21 | PROD20     | Outdoor Sports Water Bottle | Sports      | Accessories  | Soto-Elli                    | 131.51 | 273   |  |  |  |  |  |  |  |  |
| 22 | PROD21     | Multi-Purpose Storage Box   | Home        | Storage      | Gardner-Montes               | 123.46 | 76    |  |  |  |  |  |  |  |  |
| 23 | PROD22     | Carbon Fiber Tennis Racket  | Sports      | Equipment    | Flores, Pineda and Rodriguez | 484.62 | 432   |  |  |  |  |  |  |  |  |
| 24 | PROD23     | Men's Formal Shirt          | Clothing    | Tops         | Moon, Mora and Santos        | 254.04 | 360   |  |  |  |  |  |  |  |  |
| 25 | PROD24     | Foldable Camping Chair      | Sports      | Outdoor      | Cook Ltd                     | 166.06 | 340   |  |  |  |  |  |  |  |  |
| 26 | PROD25     | Women's Scarf Set           | Clothing    | Accessories  | Brown and Sons               | 174.02 | 268   |  |  |  |  |  |  |  |  |
| 27 | PROD26     | Wireless Bluetooth Speaker  | Electronics | Audio        | Reed-Schmidt                 | 322.41 | 213   |  |  |  |  |  |  |  |  |
| 28 | PROD27     | Decorative Floor Vase       | Home        | Decor        | Khan, Jones and Paul         | 254.04 | 378   |  |  |  |  |  |  |  |  |
| 29 | PROD28     | Sports Stopwatch Timer      | Sports      | Accessories  | Burns-Hansen                 | 38.76  | 137   |  |  |  |  |  |  |  |  |
| 30 | PROD29     | Fabric Sofa Throw Blanket   | Home        | Decor        | Hoffman Group                | 218.48 | 276   |  |  |  |  |  |  |  |  |

Cost IF(ISBLANK([@Cost]),AVERAGE(\$F\$2:\$F\$101),[@Cost])  
Stock IF(ISBLANK([@Stock]),AVERAGE(\$G\$2:\$G\$101),[@Stock])

Sum if the category is electronics 6647.66 SUMIF(Product\_table[Category], "Electronics", Product\_table[Cost])

| Customer_ID | Name               | Age | Gender | City              | State          | Country |
|-------------|--------------------|-----|--------|-------------------|----------------|---------|
| CUST1000    | Sheena Bonilla     | 55  | Male   | Donaldton         | North Carolina | USA     |
| CUST1001    | Eugene Nelson      | 26  | Female | Phillipsmouth     | North Dakota   | USA     |
| CUST1002    | Dr. Benjamin Price | 29  | Female | Kylebury          | Alabama        | USA     |
| CUST1003    | Jeremy Harris      | 59  | Female | Thomastown        | Ohio           | USA     |
| CUST1004    | Caroline Collins   | 64  | Female | Nicholashire      | California     | USA     |
| CUST1005    | Christine Vaughn   | 24  | Male   | East Amandamouth  | Oregon         | USA     |
| CUST1006    | Amber Jackson      | 29  | Female | Deborahborough    | South Dakota   | USA     |
| CUST1007    | Daniel Boyd        | 29  | Female | West Pamelfurt    | West Virginia  | USA     |
| CUST1008    | Melissa Howard     | 53  | Female | Maynardton        | New Hampshire  | USA     |
| CUST1009    | Ms. Maria Leach    | 64  | Male   | Melissashire      | Pennsylvania   | USA     |
| CUST1010    | Dominic Page       | 53  | Male   | Port Michaelburgh | Pennsylvania   | USA     |
| CUST1011    | Brian Lopez        | 58  | Female | East Krista       | New York       | USA     |
| CUST1012    | Kenneth Lambert    | 55  | Female | North Kathryn     | Kansas         | USA     |
| CUST1013    | Nathan Hernandez   | 32  | Female | New Becky         | Montana        | USA     |
| CUST1014    | Ashley Cohen       | 53  | Female | Lake Kristamouth  | Nebraska       | USA     |
| CUST1015    | Laura Alvarez      | 49  | Female | Lawsonhaven       | Washington     | USA     |
| CUST1016    | John Scott         | 24  | Female | South Christopher | Pennsylvania   | USA     |
| CUST1017    | Adam Cruz          | 60  | Female | Brandonville      | Georgia        | USA     |
| CUST1018    | Rhonda Moss        | 43  | Male   | South William     | Kansas         | USA     |
| CUST1019    | Michael Torres     | 46  | Female | Gonzalezfort      | New Jersey     | USA     |
| CUST1020    | Bridget Allen      | 59  | Male   | Lake Shaneburgh   | Delaware       | USA     |
| CUST1021    | Jeff Martinez      | 36  | Female | Smithburgh        | Oklahoma       | USA     |
| CUST1022    | Paul Fox           | 33  | Female | Michaelton        | Hawaii         | USA     |
| CUST1023    | Jonathan Steele    | 52  | Female | Molnabury         | Oregon         | USA     |
| CUST1024    | Heather Collins    | 40  | Male   | Rogerschester     | Idaho          | USA     |

#### Data cleaning

Customer\_ID Replacing C- with CUST

Find and Replace

Find what: C- Replace with: CUST

Match case: ☐ Match whole cell contents: ☐ Search by: Formulas

Replace all

Name: PROPER(TRIM(CLEAN(B2)))  
City: PROPER(TRIM(CLEAN(E2)))  
State: PROPER(TRIM(CLEAN(F2)))  
Country: Replacing United States of America to USA

| Customer_ID | Name               | Age | Gender | City               | State        | Country |
|-------------|--------------------|-----|--------|--------------------|--------------|---------|
| CUST1018    | Rhonda Moss        | 43  | Male   | South William      | Kansas       | USA     |
| CUST1019    | Michael Torres     | 46  | Female | Gonzalezfort       | New Jersey   | USA     |
| CUST1020    | Bridget Allen      | 59  | Male   | Lake Shaneburgh    | Delaware     | USA     |
| CUST1021    | Jeff Martinez      | 36  | Female | Smithburgh         | Oklahoma     | USA     |
| CUST1022    | Paul Fox           | 33  | Female | Michaelton         | Hawaii       | USA     |
| CUST1023    | Jonathan Steele    | 52  | Female | Molnabury          | Oregon       | USA     |
| CUST1024    | Heather Collins    | 40  | Male   | Rogerschester      | Idaho        | USA     |
| CUST1025    | Diana Taylor       | 59  | Female | Vasqueztown        | California   | USA     |
| CUST1026    | Amber Carlson      | 18  | Female | East Michellemouth | Missouri     | USA     |
| CUST1027    | Leslie Johnson     | 31  | Male   | Mollyshire         | Utah         | USA     |
| CUST1028    | Joshua Ryan        | 27  | Female | New David          | New Jersey   | USA     |
| CUST1029    | David Arnold       | 54  | Female | East Jennifer      | California   | USA     |
| CUST1030    | Jessica Acevedo    | 64  | Female | Amybury            | Pennsylvania | USA     |
| CUST1031    | Sarah Vance        | 41  | Male   | Johnsonville       | Oregon       | USA     |
| CUST1032    | Mike Jennings      | 48  | Female | Fostermouth        | Arizona      | USA     |
| CUST1033    | Robert Jones Dds   | 25  | Male   | Tristanton         | Oklahoma     | USA     |
| CUST1034    | Christopher Olson  | 27  | Female | West Richard       | Maryland     | USA     |
| CUST1035    | Rachel Myers       | 52  | Male   | Robertberg         | North Dakota | USA     |
| CUST1036    | Thomas Phillips    | 20  | Female | Lustford           | Georgia      | USA     |
| CUST1037    | Christopher Carter | 27  | Female | Brendastad         | South Dakota | USA     |
| CUST1038    | Deanna Simpson     | 35  | Female | Brownport          | North Dakota | USA     |
| CUST1039    | Tonya Reynolds     | 55  | Female | Darleneton         | Michigan     | USA     |
| CUST1040    | Eric Vega          | 50  | Male   | Williamsonland     | Kansas       | USA     |
| CUST1041    | James Gibbs        | 21  | Male   | Mooneyshire        | Alaska       | USA     |
| CUST1042    | William Martin     | 19  | Male   | Ryanburgh          | Connecticut  | USA     |
| CUST1043    | Joseph Snow        | 45  | Male   | South Michaelfort  | Vermont      | USA     |

Name: PROPER(TRIM(CLEAN(B2)))  
City: PROPER(TRIM(CLEAN(E2)))  
State: PROPER(TRIM(CLEAN(F2)))  
Country: Replacing United States of America to USA

Find and Replace

Find what: United States of America Replace with: USA

Match case: ☐ Match whole cell contents: ☐ Search by: Formulas

Replace all

Count if the state is virginia 11 COUNTIF(customer\_table[State],"Virginia")

## 4.Merging all the sheets

Customer name

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Age

XLOOKUP([@[Customer\_ID]],Customer\_table[Customer\_ID],Customer\_table[Age],FALSE)

Gender

XLOOKUP([@[Customer\_ID]],Customer\_table[Customer\_ID],Customer\_table[Gender],FALSE)

City

XLOOKUP([@[Customer\_ID]],Customer\_table[Customer\_ID],Customer\_table[City],FALSE)

State

XLOOKUP([@[Customer\_ID]],Customer\_table[Customer\_ID],Customer\_table[State],FALSE)

Country

XLOOKUP([@[Customer\_ID]],Customer\_table[Customer\_ID],Customer\_table[Country],FALSE)

Product name

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Category

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Sub category

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Brand

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Cost

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Stock

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Store name

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Store region

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Store city

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Store type

=XLOOKUP([@[Store\_ID]],Store\_table[Store\_ID],Store\_table[Store\_Type],FALSE)

Z2    =XLOOKUP([@[Store\_ID]],Store\_table[Store\_ID],Store\_table[Store\_Type],FALSE)

|    | A        | B          | C           | D          | E        | F        | G          | H        | I            | J            | K                   | L   | M      |
|----|----------|------------|-------------|------------|----------|----------|------------|----------|--------------|--------------|---------------------|-----|--------|
| 1  | Sales_ID | Order_Date | Customer_ID | Product_ID | Store_ID | Quantity | Unit_Price | Discount | Payment_Type | Total_Amount | Customer Name       | Age | Gender |
| 2  | SALE0    | 07-07-2024 | CUST1409    | PROD97     | STORE3   | 1        | 218.04     | 0.09     | Credit Card  | 217.95       | Sarah Smith         | 55  | Male   |
| 3  | SALE1    | 25-12-2023 | CUST1467    | PROD74     | STORE17  | 3        | 104.47     | 0.27     | PayPal       | 313.14       | Michael Foley       | 63  | Female |
| 4  | SALE2    | 03-06-2024 | CUST1056    | PROD49     | STORE8   | 4        | 914.05     | 0.25     | Credit Card  | 3655.95      | Christopher Sutton  | 49  | Female |
| 5  | SALE3    | 29-11-2023 | CUST1346    | PROD76     | STORE8   | 1        | 921.51     | 0.16     | COD          | 921.35       | John Jordan         | 58  | Female |
| 6  | SALE4    | 06-01-2024 | CUST1493    | PROD59     | STORE5   | 3        | 745.30     | 0.18     | Credit Card  | 2235.72      | Amanda Powell       | 28  | Male   |
| 7  | SALE5    | 26-07-2024 | CUST1116    | PROD15     | STORE7   | 3        | 147.01     | 0.19     | COD          | 440.84       | Anthony Phillips    | 18  | Female |
| 8  | SALE6    | 12-07-2025 | CUST1016    | PROD56     | STORE4   | 2        | 785.31     | 0.15     | PayPal       | 1570.47      | John Scott          | 24  | Female |
| 9  | SALE7    | 18-11-2023 | CUST1152    | PROD70     | STORE8   | 4        | 342.39     | 0.23     | PayPal       | 1369.33      | Patrick Roberts     | 41  | Female |
| 10 | SALE8    | 04-05-2025 | CUST1187    | PROD99     | STORE5   | 2        | 448.25     | 0.07     | PayPal       | 1111.68      | Dr. Suzanne Bullock | 60  | Female |
| 11 | SALE9    | 31-07-2025 | CUST1464    | PROD33     | STORE0   | 2        | 873.17     | 0.23     | PayPal       | 1746.11      | David Pierce        | 45  | Female |
| 12 | SALE10   | 13-02-2024 | CUST1065    | PROD24     | STORE14  | 1        | 278.63     | 0.07     | Credit Card  | 278.56       | Michael Thomas      | 21  | Female |
| 13 | SALE11   | 24-05-2025 | CUST1320    | PROD39     | STORE1   | 4        | 286.92     | 0.17     | Credit Card  | 1147.51      | Jessica Gonzalez    | 34  | Female |
| 14 | SALE12   | 24-08-2024 | CUST1237    | PROD88     | STORE4   | 4        | 644.58     | 0.01     | Credit Card  | 2578.31      | Wesley Short        | 41  | Female |
| 15 | SALE13   | 06-08-2025 | CUST1443    | PROD34     | STORE16  | 2        | 316.90     | 0.30     | PayPal       | 633.50       | Adam Castro         | 51  | Male   |
| 16 | SALE14   | 21-06-2025 | CUST1348    | PROD24     | STORE15  | 4        | 403.47     | 0.29     | PayPal       | 1613.59      | Sherry Cabrera      | 55  | Male   |
| 17 | SALE15   | 10-12-2024 | CUST1435    | PROD36     | STORE9   | 2        | 871.34     | 0.04     | COD          | 2161.06      | Lindsay Beasley     | 37  | Female |
| 18 | SALE16   | 13-09-2025 | CUST1276    | PROD65     | STORE19  | 2        | 451.98     | 0.25     | Credit Card  | 903.71       | Arthur Warren       | 38  | Male   |
| 19 | SALE17   | 11-12-2024 | CUST1079    | PROD27     | STORE4   | 2        | 502.69     | 0.23     | Credit Card  | 1005.15      | Molly Gardner       | 22  | Female |
| 20 | SALE18   | 13-08-2024 | CUST1094    | PROD39     | STORE4   | 2        | 559.23     | 0.25     | COD          | 1118.21      | David Patterson     | 43  | Female |
| 21 | SALE19   | 18-03-2024 | CUST1239    | PROD60     | STORE13  | 4        | 513.98     | 0.04     | COD          | 2055.90      | Maria Hill          | 58  | Female |
| 22 | SALE20   | 05-07-2025 | CUST1144    | PROD72     | STORE17  | 3        | 986.04     | 0.27     | COD          | 2957.85      | Justin Cross        | 62  | Female |

merged table   Customer\_Dim   Product\_Dim   Store\_Dim   Sales\_Fact   Customer ...   89%

22     $\text{=XLOOKUP}([@[Store\_ID]],Store\_table[Store\_ID],Store\_table[Store\_Type],FALSE)$

|    | N                    | O              | P     | Q                            | R                | S            | T                             | U      | V     | W                              |
|----|----------------------|----------------|-------|------------------------------|------------------|--------------|-------------------------------|--------|-------|--------------------------------|
|    | Customer city        | Customer state | Count | Product name                 | Product category | Sub category | Brand                         | cost   | Stocd | Store name                     |
| 1  | Kathleenport         | Wyoming        | USA   | Cricket Bat                  | Sports           | Team Sports  | Khan LLC                      | 245.98 | 471   | Johnson-Wright                 |
| 2  | Oneillfurt           | Vermont        | USA   | Bluetooth Car Audio System   | Electronics      | Audio        | Bruce-Bowen                   | 403.71 | 159   | Holloway Ltd                   |
| 3  | Deborahberg          | Pennsylvania   | USA   | Men's Roll-On Perfume        | Beauty           | Fragrance    | Sullivan and Sons             | 43.29  | 267   | Bean, Jones and Park           |
| 4  | Port Crystalberg     | Tennessee      | USA   | Wireless Mouse               | Electronics      | Computer     | Larson Ltd                    | 392.89 | 320   | Bean, Jones and Park           |
| 5  | South William        | Alabama        | USA   | Men's Hair Styling Gel       | Beauty           | Haircare     | Cline-Mccarthy                | 17.31  | 399   | Gray Ltd                       |
| 6  | North Davidport      | Rhode Island   | USA   | 4K Home Projector            | Electronics      | Appliances   | Thomas LLC                    | 465.34 | 301   | Martinez-Mullins               |
| 7  | South Christopher    | Pennsylvania   | USA   | Premium Blanket Throw        | Home             | Decor        | Hernandez, Kennedy and Porter | 461.47 | 190   | Alexander Group                |
| 8  | Jaredville           | Missouri       | USA   | Kitchen Storage Rack         | Home             | Storage      | Duran Group                   | 138.05 | 89    | Bean, Jones and Park           |
| 9  | Lake Blaketon        | Alabama        | USA   | Stainless Steel Cookware Set | Home             | Kitchen      | Campbell Ltd                  | 201.41 | 146   | Gray Ltd                       |
| 10 | Grayland             | Washington     | USA   | Modern Wall Clock            | Home             | Decor        | Weiss-Watson                  | 179.77 | 305   | Rivera-Warner                  |
| 11 | Woodmouth            | Nevada         | USA   | Foldable Camping Chair       | Sports           | Outdoor      | Cook Ltd                      | 166.06 | 340   | Mcdonald Inc                   |
| 12 | South Darrenland     | Iowa           | USA   | Portable Power Bank          | Electronics      | Mobile       | Cummings and Sons             | 326.06 | 301   | Martinez, Adkins and Johnson   |
| 13 | Susanstad            | Montana        | USA   | Sports Wrist Sweatbands      | Sports           | Accessories  | Olsen-Daniels                 | 131.24 | 232   | Alexander Group                |
| 14 | New Donnabury        | Texas          | USA   | Gaming Mechanical Keyboard   | Electronics      | Computer     | Lucero LLC                    | 419.28 | 86    | Frederick-Wright               |
| 15 | Shannonview          | Pennsylvania   | USA   | Foldable Camping Chair       | Sports           | Outdoor      | Cook Ltd                      | 166.06 | 340   | Shelton PLC                    |
| 16 | Wesleyhaven          | Idaho          | USA   | Women's Summer Dress         | Clothing         | Tops         | Smith-Mills                   | 223.43 | 154   | Moore, Padilla and Butler      |
| 17 | Erichshire           | Nevada         | USA   | Adjustable Dumbbells 20kg    | Sports           | Fitness      | Montgomery, Benton and Miller | 240.63 | 423   | Mann-Austin                    |
| 18 | Nicoleborough        | Arkansas       | USA   | Decorative Floor Vase        | Home             | Decor        | Khan, Jones and Paul          | 254.04 | 378   | Alexander Group                |
| 19 | New Amyport          | Nevada         | USA   | Portable Power Bank          | Electronics      | Mobile       | Cummings and Sons             | 326.06 | 301   | Alexander Group                |
| 20 | Zamoraborough        | Nevada         | USA   | Luxury Perfume Spray         | Beauty           | Fragrance    | Gibson, Green and Lopez       | 482.75 | 444   | Gardner, Humphrey and Contrera |
| 21 | North Anthonychester | Nevada         | USA   | Professional Jump Rope       | Sports           | Fitness      | Davis PLC                     | 477.61 | 300   | Holloway Ltd                   |

merged table    Customer\_Dim    Product\_Dim    Store\_Dim    Sales\_Fact    Customer ...    +    :    <

22     $\text{=XLOOKUP}([@[Store\_ID]],Store\_table[Store\_ID],Store\_table[Store\_Type],FALSE)$

|    | T                             | U      | V     | W                               | X            | Y                  | Z          | AA | AB | AC | AD | AE | AF |
|----|-------------------------------|--------|-------|---------------------------------|--------------|--------------------|------------|----|----|----|----|----|----|
|    | Brand                         | cost   | Stocd | Store name                      | Store region | Store city         | store type |    |    |    |    |    |    |
| 1  | Khan LLC                      | 245.98 | 471   | Johnson-Wright                  | North        | Hoodberg           | Outlet     |    |    |    |    |    |    |
| 2  | Bruce-Bowen                   | 403.71 | 159   | Holloway Ltd                    | North        | Bensonhaven        | Flagship   |    |    |    |    |    |    |
| 3  | Sullivan and Sons             | 43.29  | 267   | Bean, Jones and Park            | West         | Thompsonborough    | Online     |    |    |    |    |    |    |
| 4  | Larson Ltd                    | 392.89 | 320   | Bean, Jones and Park            | West         | Thompsonborough    | Online     |    |    |    |    |    |    |
| 5  | Cline-Mccarthy                | 17.31  | 399   | Gray Ltd                        | North        | New Michaelchester | Outlet     |    |    |    |    |    |    |
| 6  | Thomas LLC                    | 465.34 | 301   | Martinez-Mullins                | South        | Williamsfurt       | Online     |    |    |    |    |    |    |
| 7  | Hernandez, Kennedy and Porter | 461.47 | 190   | Alexander Group                 | South        | Port Sharon        | Online     |    |    |    |    |    |    |
| 8  | Duran Group                   | 138.05 | 89    | Bean, Jones and Park            | West         | Thompsonborough    | Online     |    |    |    |    |    |    |
| 9  | Campbell Ltd                  | 201.41 | 146   | Gray Ltd                        | North        | New Michaelchester | Outlet     |    |    |    |    |    |    |
| 10 | Weiss-Watson                  | 179.77 | 305   | Rivera-Warner                   | East         | Washingtonfurt     | Flagship   |    |    |    |    |    |    |
| 11 | Cook Ltd                      | 166.06 | 340   | Mcdonald Inc                    | South        | Christopherland    | Outlet     |    |    |    |    |    |    |
| 12 | Cummings and Sons             | 326.06 | 301   | Martinez, Adkins and Johnson    | North        | Danielmouth        | Flagship   |    |    |    |    |    |    |
| 13 | Olsen-Daniels                 | 131.24 | 232   | Alexander Group                 | South        | Port Sharon        | Online     |    |    |    |    |    |    |
| 14 | Lucero LLC                    | 419.28 | 86    | Frederick-Wright                | West         | New Jimmymouth     | Flagship   |    |    |    |    |    |    |
| 15 | Cook Ltd                      | 166.06 | 340   | Shelton PLC                     | North        | East Jenniferstad  | Flagship   |    |    |    |    |    |    |
| 16 | Smith-Mills                   | 223.43 | 154   | Moore, Padilla and Butler       | South        | East Johnstad      | Flagship   |    |    |    |    |    |    |
| 17 | Montgomery, Benton and Miller | 240.63 | 423   | Mann-Austin                     | East         | Vasquezborough     | Online     |    |    |    |    |    |    |
| 18 | Khan, Jones and Paul          | 254.04 | 378   | Alexander Group                 | South        | Port Sharon        | Online     |    |    |    |    |    |    |
| 19 | Cummings and Sons             | 326.06 | 301   | Alexander Group                 | South        | Port Sharon        | Online     |    |    |    |    |    |    |
| 20 | Gibson, Green and Lopez       | 482.75 | 444   | Gardner, Humphrey and Contreras | East         | Johnsonville       | Flagship   |    |    |    |    |    |    |
| 21 | Davis PLC                     | 477.61 | 300   | Holloway Ltd                    | North        | Bensonhaven        | Flagship   |    |    |    |    |    |    |

merged table    Customer\_Dim    Product\_Dim    Store\_Dim    Sales\_Fact    Customer ...    +    :    <

## 5.Questions to create pivot table

### Product Analysis

- Which product category generates the most sales?
- Which sub-category performs best?
- Which products are the best sellers?
- Which products generate the highest revenue?

### Customer Analysis

- Which gender contributes more to sales?
- Which cities have the highest sales?
- Which customers generate the most revenue?

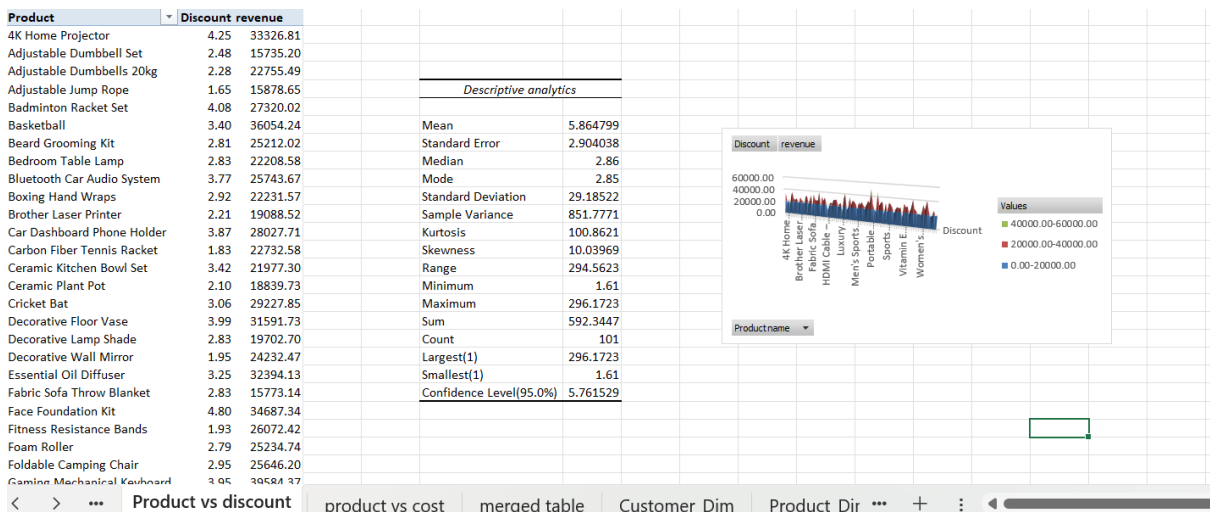
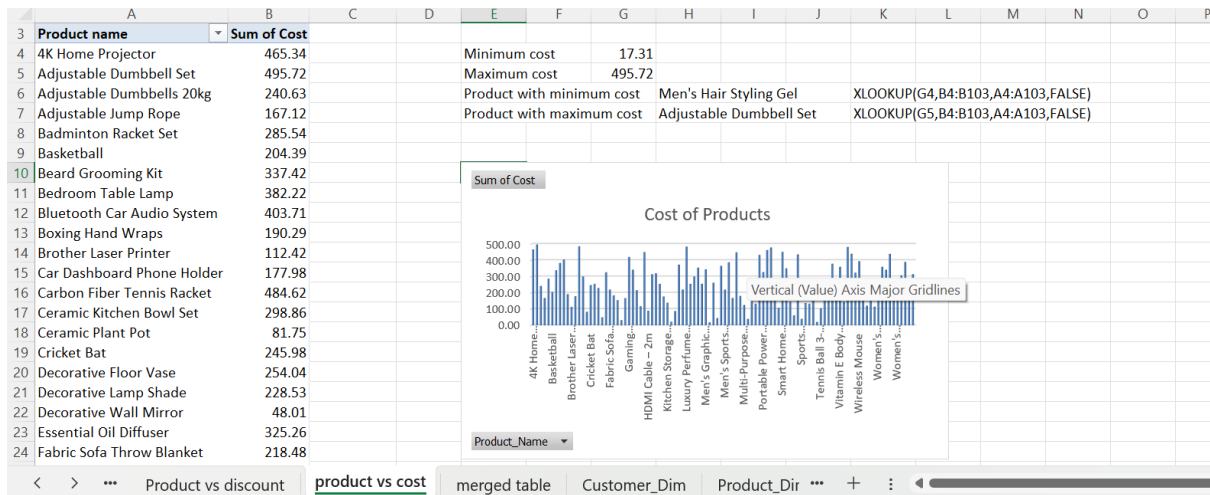
### Store Analysis

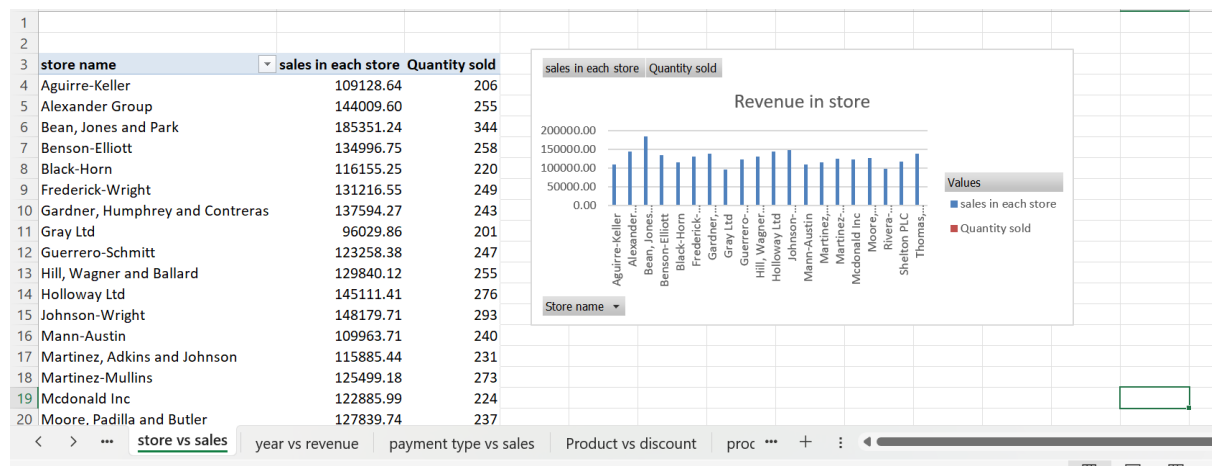
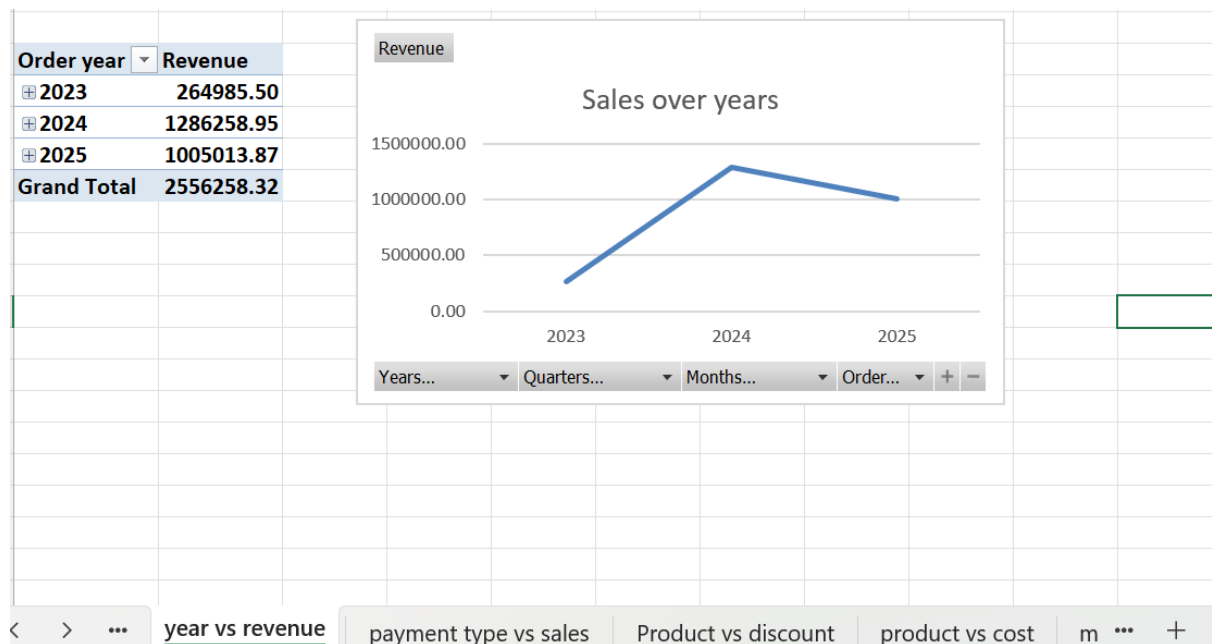
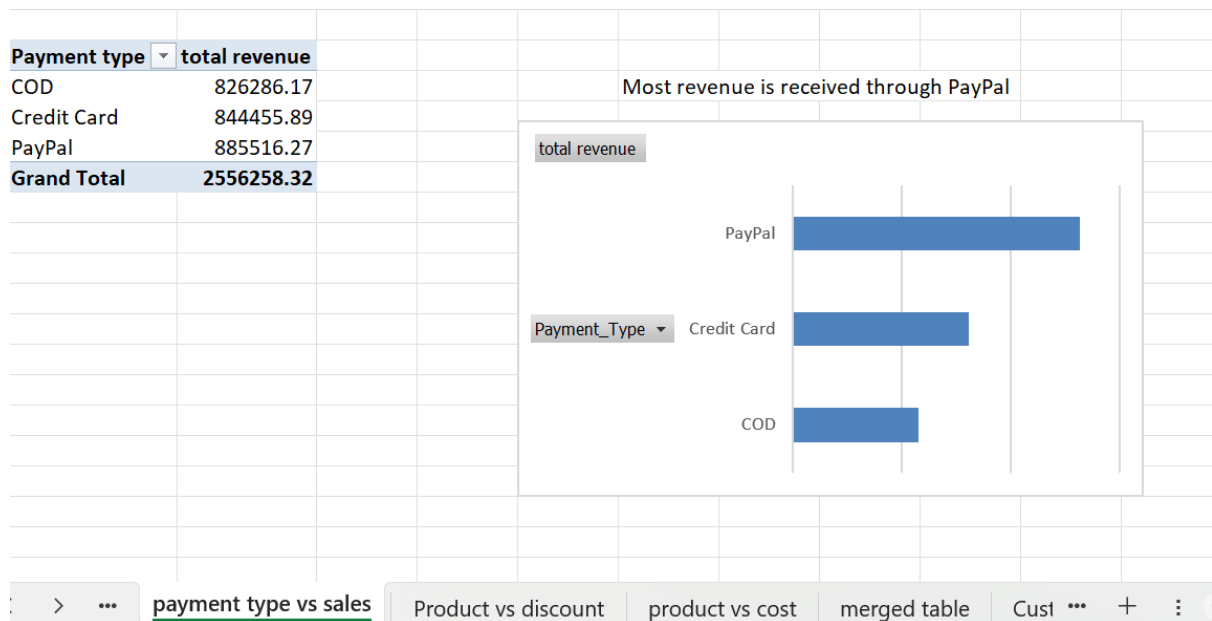
- Which store generates the highest revenue?
- Which region performs best?

- Which store sells the highest quantity?

## Sales Analysis

- How many orders were placed?
- How are sales changing over time?
- Which payment method generates the highest sales?
- How much sales are generated at different discount levels?





< > ... Customer vs gender city vs sales Customer vs sales ... + :

< > ... Customer vs gender city vs sales Customer vs sales region vs store store vs sa ... + :

A bar chart titled 'Count of Customer\_ID' and 'Average of Age'. The x-axis is labeled 'Gender' with categories 'Female' and 'Male'. The y-axis is labeled 'Values' and ranges from 0 to 300. For each gender, there are two bars: a blue bar for 'Count of Customer\_ID' and a red bar for 'Average of Age'. The blue bars are significantly taller than the red bars.

| Gender | Count of Customer_ID | Average of Age |
|--------|----------------------|----------------|
| Female | 260                  | 40             |
| Male   | 230                  | 40             |

|    | A                  | B                    | C | D               | E         | F      | G | H | I | J | K |
|----|--------------------|----------------------|---|-----------------|-----------|--------|---|---|---|---|---|
| 3  | <b>Category</b>    | <b>Total Revenue</b> |   |                 |           |        |   |   |   |   |   |
| 4  | <b>Beauty</b>      | <b>381526.08</b>     |   | Maximum revenue | 381526.08 | Sports |   |   |   |   |   |
| 5  | Fragrance          | 54923.20             |   | Minimum revenue | 25212.02  | Beauty |   |   |   |   |   |
| 6  | Grooming           | 25212.02             |   |                 |           |        |   |   |   |   |   |
| 7  | Haircare           | 119336.10            |   |                 |           |        |   |   |   |   |   |
| 8  | Makeup             | 75741.81             |   |                 |           |        |   |   |   |   |   |
| 9  | Skincare           | 106312.95            |   |                 |           |        |   |   |   |   |   |
| 10 | <b>Clothing</b>    | <b>442751.50</b>     |   |                 |           |        |   |   |   |   |   |
| 11 | Accessories        | 94596.46             |   |                 |           |        |   |   |   |   |   |
| 12 | Bottoms            | 79182.55             |   |                 |           |        |   |   |   |   |   |
| 13 | Footwear           | 107211.91            |   |                 |           |        |   |   |   |   |   |
| 14 | Outerwear          | 110137.71            |   |                 |           |        |   |   |   |   |   |
| 15 | Tops               | 51622.87             |   |                 |           |        |   |   |   |   |   |
| 16 | <b>Electronics</b> | <b>593563.45</b>     |   |                 |           |        |   |   |   |   |   |
| 17 | Accessories        | 176880.50            |   |                 |           |        |   |   |   |   |   |
| 18 | Appliances         | 62386.75             |   |                 |           |        |   |   |   |   |   |
| 19 | Audio              | 121741.16            |   |                 |           |        |   |   |   |   |   |
| 20 | Computer           | 181939.69            |   |                 |           |        |   |   |   |   |   |
| 21 | Mobile             | 50615.34             |   |                 |           |        |   |   |   |   |   |
| 22 | <b>Home</b>        | <b>502977.27</b>     |   |                 |           |        |   |   |   |   |   |

Total Revenue

Total revenue based on Product

| Product Category | Sub-Category | Revenue    |
|------------------|--------------|------------|
| Beauty           | Fragrance    | 54,923.20  |
|                  | Haircare     | 119,336.10 |
|                  | Skincare     | 106,312.95 |
|                  | Bottoms      | 79,182.55  |
|                  | Outerwear    | 110,137.71 |
| Clothing         | Accessories  | 94,596.46  |
|                  | Audio        | 121,741.16 |
|                  | Mobile       | 50,615.34  |
|                  | Decor        | 181,939.69 |
|                  | Kitchen      | 62,386.75  |
| Electronics      | Storage      | 51,622.87  |
|                  | Equipment    | 176,880.50 |
|                  | Outdoor      | 110,137.71 |
|                  | Computer     | 181,939.69 |
|                  | Appliances   | 62,386.75  |
| Home             | Accessories  | 176,880.50 |
|                  | Audio        | 121,741.16 |
|                  | Mobile       | 50,615.34  |
|                  | Decor        | 181,939.69 |
|                  | Kitchen      | 62,386.75  |
| Sports           | Storage      | 51,622.87  |
|                  | Equipment    | 176,880.50 |
|                  | Outdoor      | 110,137.71 |
|                  | Computer     | 181,939.69 |
|                  | Appliances   | 62,386.75  |

| Count of Store_ID | Count of Store_ID2 | Forecast(Count of Store_ID) | Lower Confidence Bound(Count of Store_ID) | Upper Confidence Bound(Count of Store_ID) |
|-------------------|--------------------|-----------------------------|-------------------------------------------|-------------------------------------------|
| 2                 | 2                  |                             |                                           |                                           |
| 6                 | 6                  |                             |                                           |                                           |
| 10                |                    |                             |                                           |                                           |
| 14                |                    |                             |                                           |                                           |
| 18                |                    |                             |                                           |                                           |
| 22                |                    |                             |                                           |                                           |
| 26                |                    |                             |                                           |                                           |
| 30                |                    |                             |                                           |                                           |
| 32                |                    |                             |                                           |                                           |

Forecast for region vs store

Category vs sales   product vs sales   Customer vs gender

## Sample relationship

| Status | Table ▲                   | Related Lookup Table         |
|--------|---------------------------|------------------------------|
| Active | total_table (Customer_ID) | Customer_table (Customer_ID) |
| Active | total_table (Product_ID)  | Product_table (Product_ID)   |

New...  
Auto-Detect...  
Edit...  
Activate  
Deactivate  
Delete

Close

## Dashboard:

